

JESSICA QIU

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EDUCATION

Boston University College of Engineering

May 2027

Bachelor of Science in Electrical Engineering, Minor in Biomedical Engineering

GPA: 3.8

Activities & Societies: Theta Tau Professional Engineering Fraternity (Academic Chair); BU Mars Rover Club (Electrical and Life Detection Team); BU Society of Asian Scientists and Engineers (Mentor).

EXPERIENCE

Undergraduate Research Assistant

May 2026 – July 2026

Centre for Bio-Devices and Bio-Informatics, Nanyang Technological University

Singapore

- Assisted in the fabrication and characterization of polymer microfibers for a bio-optical sensing platform, working closely with PhD researchers to optimize fabrication processes and support device development.
- Performed optical microscopy and image analysis to evaluate microfiber morphology, dimensional consistency, and structural quality, generating quantitative data to guide experimental design and process refinement.
- Supported investigations of microfiber lasing through optical characterization and experimental testing, contributing to the development of non-invasive pathogen monitoring technologies for breath diagnostics.

Undergraduate Research Assistant

Nov 2024 – Present

BU Ji Xin-Cheng Research Group

Boston, MA

- Assisted in the development of advanced nonlinear vibrational imaging platforms (VREF and SRP microscopy) through optical system design, component integration, laser alignment, calibration, and validation, collaborating with postdoctoral researchers to improve imaging performance, collect biological data, and experimental reproducibility.
- Applied fluorescence microscopy and quantitative image analysis to characterize lysosome-related organelles, processing spectroscopic and biological datasets to investigate cellular aging and stress-response mechanisms.
- Collaborated with biologists to culture *C. elegans* experimental models, supporting interdisciplinary microscopy studies that integrated optical measurements, quantitative imaging workflows, and biological assay development.

Undergraduate Teaching Assistant - ENG BE/EC 556: Optical Spectroscopic Imaging

Jan 2025 – Present

BU College of Engineering

Boston, MA

- Led weekly office hours and provided support for graduate-level coursework in optical spectroscopy and imaging, helping students develop stronger foundations in imaging physics and spectroscopy.
- Evaluated assignments, exams, and projects, providing detailed feedback to reinforce key concepts in spectroscopic imaging and microscopy, while collaborating with the instructor on course administration and instructional support.

Software Engineering Intern I

June 2024 – August 2025

Pulp Internet Corporation

Brooklyn, NY

- Developed Data Stories templates using JavaScript, Python, and data visualization tools, transforming transcript datasets into interactive narratives with dynamic charts, speaker insights, and scroll-driven storytelling elements.
- Contributed to Persona Engine development by integrating OpenAI workflows and NLP techniques to analyze communication patterns, extract key themes, and generate audience-specific insights from large transcript datasets.
- Built automated data processing and visualization pipelines using Python, pandas, NumPy, and Streamlit, reducing manual dataset preparation and enabling researchers to explore rhetorical dynamics and persuasion strategies.

PROJECTS

Biometric Sensor for Exercise Device

Oct – Dec 2024

- Led a 4-person team in the design of a wearable biometric sensing device that monitored wrist acceleration and rotation during tennis swings, providing real-time haptic feedback when user-defined performance thresholds were met.
- Designed and integrated the device's electronic hardware, including sensor, power, and haptic-feedback circuitry, using an Arduino Nano, IMU, OLED display, and rotary encoder to enable real-time motion tracking and user interaction.
- Developed firmware to process acceleration and gyroscope data, support configurable threshold settings, and achieve a $\pm 5\%$ measurement accuracy target through calibration, testing, and iterative hardware-software refinement.

SKILLS

Hardware: Oscilloscopes, Multimeters, Function Generators, DC Power Supplies, Soldering, PCB Design, Breadboarding, Arduino, Microcontrollers, Optical Alignment, Fluorescence Microscopy

Software Tools: KiCad, LaTeX, Microsoft Office, Revit, AutoCAD.

Programming: Python, MATLAB, Verilog, JavaScript.